

Circuit Breakers: An Interactive Cybersecurity Game Employing Adaptive Procedural Generation Techniques to Cultivate Digital Literacy and Enhance Cyber Threat Awareness

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Abstract- Cybersecurity skills are growing increasingly more important in today's rapidly digitizing world. However, cybersecurity education, particularly for students aged 8-18 faces challenges with regard to both engagement with the subject via traditional learning methods and learners' prior exposure to the topic. Circuit Breakers is a 2D action roguelike game developed to provide a fun and engaging supplement to current teaching in order to address these issues. The player must defeat hordes of enemies based on real-life life cyberthreats using weapons and items based on different cybersecurity measures. The game utilizes procedural generation and adaptive difficulty to tailor the experience for the knowledge and skill level of the player, encouraging them to learn about cybersecurity concepts they may otherwise not have known along the way through a gamified perspective. The initiative is inspired by research that shows how game-based learning can improve practice, creativity, and retention. This study validated the model through comprehensive Software Quality Assurance (SQA), confirming high functional stability (97.06% unit test passing rate) and User Acceptance Testing (UAT). UAT confirmed exceptional player engagement, evidenced by an aggregate Total Play Time of 4,180.57 minutes (approximately 70 hours) across 31 participants. Crucially, the strategic learning mechanisms proved highly effective, with players strongly agreeing to recommend the game for digital literacy learning ($x = 4.77\%$) and exhibiting a 100% Quiz Accuracy Rate among highly engaged users, confirming that knowledge acquisition is essential for game mastery. This effort converts abstract cyberthreats into engaging, consequence-based experiences, effectively counteracting educational disengagement and promoting sustained learning in cybersecurity.

Keywords – Adaptive Difficulty, Cybersecurity, Cybersecurity Education, Digital Literacy, Educational Games, Game-Based Learning, Gamification, Interactive Learning, Procedural Generation, Roguelike Games, Serious Games

I. INTRODUCTION

In today's digital era, technology, and gameplay have transformed education by providing interactive tools that boost engagement, deepen learning, encourage collaboration, and link theoretical concepts to practical scenarios. As a complementary resource, technology-enhanced educator-learner interactions foster active and meaningful educational experiences (Hébert and Jenson, 2019). Educational games, in particular, excelled as innovative platforms that blended entertainment with instruction, making them ideal for teaching intricate subjects like cybersecurity. The effectiveness of these games relied on harmonizing fun with educational value, ensuring gameplay supported clear learning goals (Chen et al., 2021).

Cybersecurity education grappled with challenges that traditional methods often failed to overcome. The 2021 Verizon Data Breach Investigations Report (DBIR) noted that human error and limited awareness drove many data breaches, underscoring the need for stronger digital literacy (Verizon, 2021). A survey of 51 potential users conducted for this study revealed that while 70.6% rated their interest in learning cybersecurity through a game-based approach as 4 or 5 on a 1–5 scale, only 39.2% showed similar enthusiasm for traditional methods like lectures and textbooks. This stark contrast

highlighted a demand for more engaging alternatives. Likewise, the NIST Special Publication 800-50 warned that poor comprehension of digital threats heightened vulnerability, pushing for engaging, proactive learning approaches (National Institute of Standards and Technology, 2003; National Institute of Standards and Technology, 2023). However, conventional teaching, often dense with theory, struggled to captivate learners, especially younger ones, and rarely bridged the divide between knowledge and real-world practice, as noted by Xiao et al. (2023). Supporting this, 68.6% of survey respondents in this study favored a roguelike game to make cybersecurity education impactful.

Circuit Breakers was a roguelike action game developed to address this issue. In Circuit Breakers, the player was tasked with controlling a character represented as a device and using various items and weapons represented as various cybersecurity measures they collected in a run to defeat enemies, represented as different cybersecurity threats. Gameplay was split into different sessions, or “runs,” which lasted until either the player was defeated or they defeated the final boss after a certain duration, with the player being encouraged to keep doing more runs in order to further upgrade their system, fill out their encyclopedia of threats and items, and unlock cosmetics. This process was facilitated by the game’s adaptive difficulty and procedural generation systems that added variance and challenge to subsequent runs. Threats, items, and weapons were based on real-life cybersecurity concepts, which encouraged the player to engage and learn more about cybersecurity to improve their performance in the game. This design aligned with game-based learning (GBL), which merged game mechanics with educational content to sharpen critical thinking and problem-solving (Tamosevicius, 2022). The survey supported this approach, with 68.6% of respondents expressing high interest in a game combining roguelike action with cybersecurity education, validating the game’s potential to captivate and educate.

Despite the increasing importance of cybersecurity education in the Philippines' educational system, there remained a significant gap in the availability of game-based learning tools that were both engaging and educationally robust, specifically tailored to the developmental needs of students aged 8–18. Current approaches, often lecture-based and theoretical, lacked the adaptability and appeal needed to sustain student interest across this diverse age range. Circuit Breakers aimed to address this gap by offering an innovative supplement to cybersecurity learning that combined interactive roguelike gameplay with practical, research-driven cybersecurity education, filling a critical void in the current educational landscape.

Traditional cybersecurity education methods, such as classroom lectures, static slides, online tutorials (e.g., PDFs or non-interactive videos), and textbook exercises, often failed to engage learners, particularly younger audiences. In cybersecurity, Van Steen and Deeleman (2021) reported that traditional training methods achieved only 33% engagement among learners, compared to 67% for gamified approaches. A study showed that there was no improvement in employees’ ability to identify phishing attacks despite all of them completing the annual cybersecurity training due to a lack of engagement (Ho et al., 2024). The survey conducted by this study of 51 potential users aligned with this, showing that only 39.2% were highly interested in traditional methods, while 70.6% preferred a game-based approach. This disengagement underscored the need for an interactive medium like Circuit Breakers to make cybersecurity education accessible and

impactful. To evaluate its effectiveness, the game used measurable outcomes like in-game performance data (e.g., challenge completion rates), embedded quizzes after key levels, and pre- and post-game tests to track learning gains and retention.

Limited exposure to cybersecurity concepts among youth stemmed from multiple root causes: a lack of cybersecurity topics in school curricula, gaps in parental knowledge or resources to teach digital safety, and insufficient engaging content in digital media. A 2024 report by Kital noted that only 19% of educational institutions in the Asia Pacific region, including the Philippines, had robust cybersecurity programs, leaving a significant gap in formal education (Kital, 2024). A 2021 National Cyber Security Alliance (NCSA) survey showed that only 13% of students aged 13–18 understood VPNs and 10% recognized multi-factor authentication, highlighting a global knowledge gap (Olmstead & Smith, 2017). The survey conducted by this study indicated that 41.2% of respondents rated their digital literacy as beginner or intermediate, supporting the existence of a significant educational gap, especially critical for the 8–18 age group targeted by Circuit Breakers. The game addressed this by using relatable analogies (e.g., a firewall as a defensive item) to deliver age-appropriate, engaging education.

There were few alternatives in learning cybersecurity other than traditional methods, such as lectures, online courses, tests, etc., and educational games often lacked depth or excitement, relying on repetitive mechanics that failed to sustain engagement or enhance long-term learning. Ho et al. (2024) conducted a study involving 19,500 employees in a healthcare organization for a span of an 8-month randomized controlled experiment, showing no correlation between annual cybersecurity training completion and ability to avoid phishing attacks, which was largely caused by a lack of engagement, indicating the need for better methods of training other than traditional methods such as lectures or tests. A 2019 meta-analysis by Clark et al. found that while 70% of educational games boosted short-term engagement, only 30% achieved significant knowledge retention after six months (Clark et al., 2019). The survey conducted by this study found that 68.6% of respondents favored a roguelike format for its replay ability and dynamic challenges, indicating a demand for modern, adaptive methods. Circuit Breakers leveraged procedural generation and adaptive mechanics to create compelling, educationally rich gameplay that improved retention through hands-on, diverse experiences.

The game placed players in a digital environment where they confronted cyber threats like malware and phishing, using in-game challenges and a "Threat Encyclopedia" to teach real-world cybersecurity concepts. This objective addressed the decline of interest in traditional methods by offering an engaging, interactive alternative. Research by Van Steen and Deeleman (2021) showed that gamified training increased engagement by 34% over traditional methods, countering learning disengagement. Ho et al. (2024) found that the results of traditional training had no improvement in employees' ability to identify phishing attacks due to a lack of engagement, supporting Circuit Breakers' objective to make learning more engaging through its roguelike repetitive game format.

Players faced scenarios requiring them to choose hardware upgrades (e.g., CPU, GPU) and cybersecurity tools (e.g., firewalls, VPNs), reinforcing proactive technology use. This objective bridged the exposure gap identified by Kital (2024), where only 19% of Asia Pacific educational institutions had robust cybersecurity programs. Wouters et al. (2013) found

that serious games improved retention by 20% over traditional methods, suggesting that Circuit Breakers could effectively teach youth through interactive challenges.

Built in Unity, the game featured procedurally generated enemy patterns and item drops for replay ability, alongside adaptive mechanics like dynamic difficulty adjustment (DDA) to balance challenge and engagement. This objective countered the retention issue in educational games with 30% retention (Clark et al., 2019), by using procedural generation to create varied scenarios. Hamari et al. (2014) found that adaptive content improved retention by 40%, aligning with Circuit Breakers' dynamic design to sustain engagement and reinforce learning over time. Ho et al. (2024) claimed the need for more interactive and engaging training methods, supporting the need for learning alternatives such as Circuit Breakers, which had the potential to make learning impactful through its roguelike format centered around creating an interactive and engaging environment.

This research provided significant benefits to the primary users, Filipino K-12 learners aged 8 to 18, by offering an interactive roguelike game that transformed abstract cybersecurity concepts into concrete, hands-on challenges. Although not the primary users, educators also stood to gain a powerful supplementary tool that complemented traditional cybersecurity instruction. Parents and guardians also benefited from Circuit Breakers' account-based progress tracking as they were able to engage in their children's learning journey by monitoring session lengths, quiz results, and topic progression gathered through a secure Firebase Authentication, encouraging positive digital habits and informed discussions about online safety.

This study also contributed to the field of educational technology by demonstrating how procedural generation, adaptive mechanics, and cloud-backed analytics could be seamlessly integrated into a serious game for scalable, low-barrier deployment. Circuit Breakers established a replicable framework for future research in serious game design, offering concrete evidence of learning gains, measured via pre- and post-tests, in-game metrics, and qualitative feedback, that could guide curriculum development and national policy on cybersecurity education. In doing so, the project not only filled a critical gap in age-appropriate digital literacy tools but also advanced our collective capacity to cultivate a security-minded generation ready to navigate an increasingly complex cyber landscape.

A. Conceptual Framework

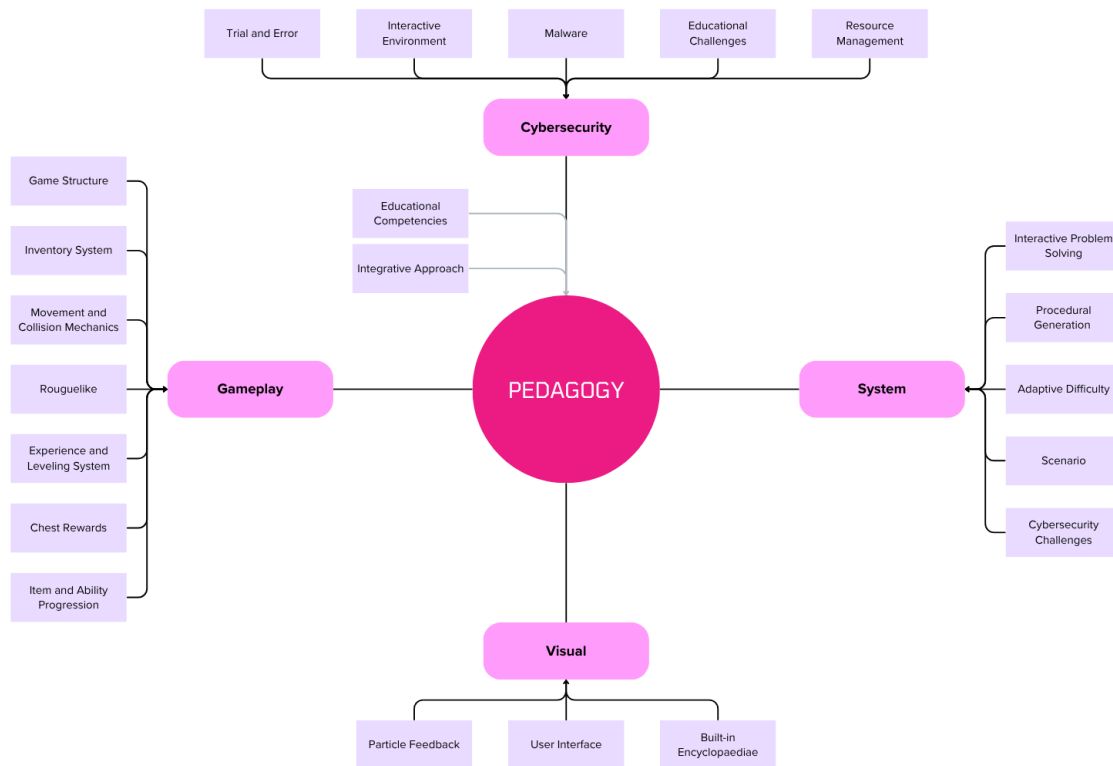


Figure 1: Balancing Pedagogy and Game Design in Cybersecurity Education

The model in Figure 1 frames Circuit Breakers as a pedagogical game built on integrative learning, combining multiple disciplines to strengthen cognitive skills, values, and practical problem-solving. Its educational design relied on four interconnected components: gameplay, which defined player interaction, progression, and mechanics; cybersecurity, which provided the core learning content through trial-and-error challenges, threat awareness, and resource management; visual design, which used engaging effects, an intuitive UI, and an in-game encyclopedia to support understanding; and system design, which governed player-environment interaction through procedural generation, adaptive difficulty, scenario building, and cybersecurity challenges. Together, these elements created an interactive framework that taught cybersecurity concepts in a dynamic, engaging, and learner-centered method.

The balance between education and fun in Circuit Breakers was achieved through an integrative pedagogical design that embedded cybersecurity concepts directly into core gameplay, visuals, and system interactions, ensuring that learning naturally emerged from exploration, progression, and problem-solving rather than feeling separate from play.

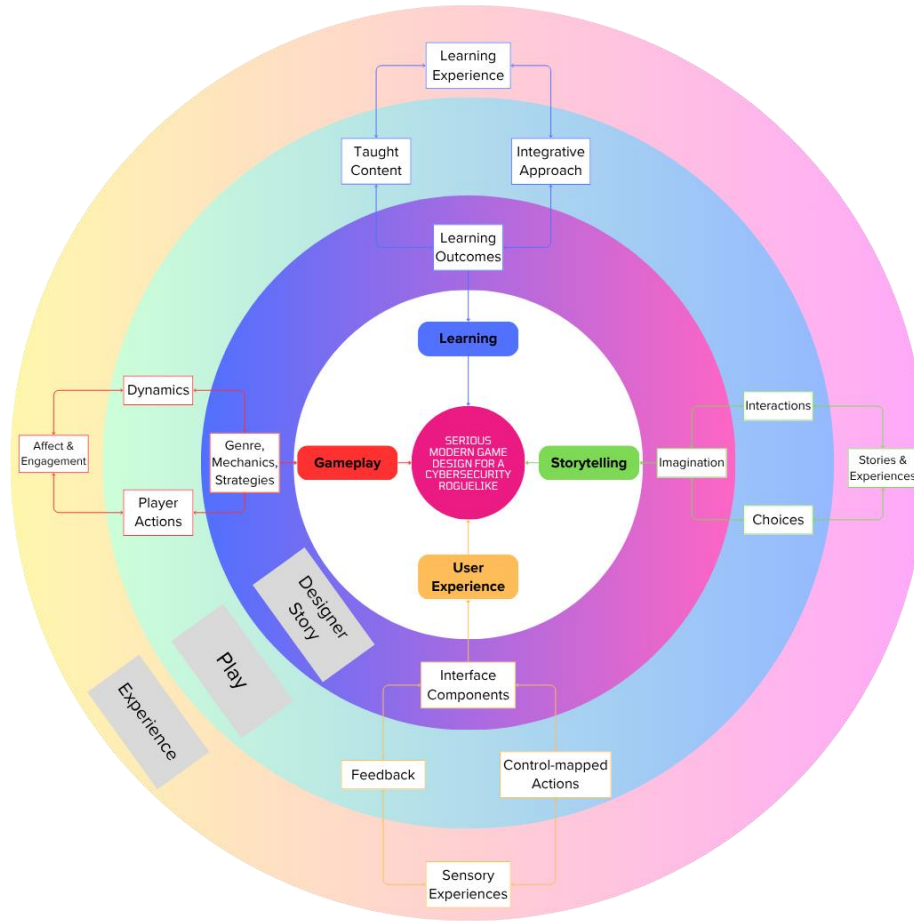


Figure 2: Serious Modern Game Design for a Cybersecurity Roguelike

Figure 2 expanded on Figure 1 by applying the Serious Modern Game Design framework, derived from Brian Winn’s Design–Play–Experience Framework, to further define Circuit Breakers’ core components. This circular model highlighted four interconnected elements: learning, storytelling, gameplay, and user experience, each shaping how players absorb content, engage with the environment, interact with mechanics, and form responses. These elements operated across three layers: the Designer’s Story (the intended design), Play (the player’s mediated interaction shaped by both design and personal background), and Experience (the unique outcomes each player derived). This framework helped evaluate how effectively the game supported learning by analyzing both game design and player behavior, using metrics such as learning outcomes, interactions, choices, sensory engagement, mechanics, strategies, and overall player experience to determine the conditions that maximized or limited educational impact.

B. Theoretical Framework

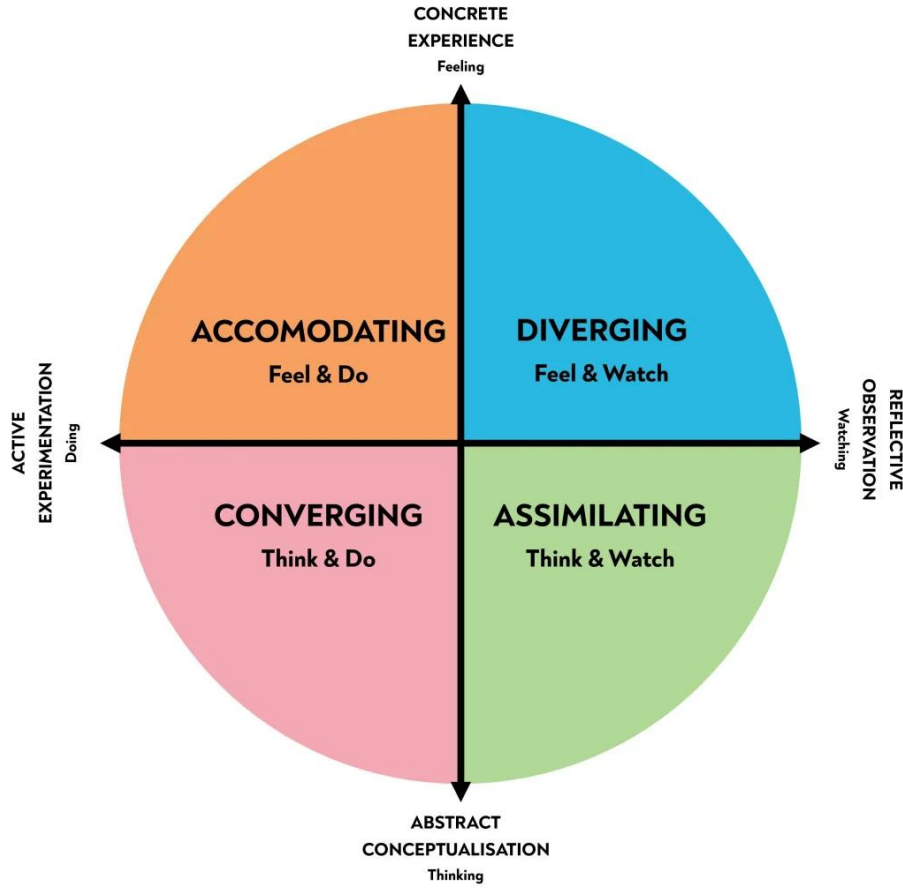


Figure 3: Kolb's Experiential Learning Theory

Figure 3 applied Kolb's Experiential Learning Theory to Circuit Breakers by framing gameplay as a four-stage learning cycle: concrete experience, reflective observation, abstract conceptualization, and active experimentation, through which players transformed in-game encounters into meaningful understanding. As players discovered items, threats, and challenges, reflected on their functions, formed strategic ideas, and tested these strategies in gameplay, the game supported continuous learning that mirrored real-world application. Kolb's four learning styles of Diverging, Assimilating, Converging, and Accommodating can be reflected in different player types, such as planner, analyst, problem solver, and challenger, allowing Circuit Breakers to identify player tendencies through quiz responses and performance data and adapt the learning experience accordingly.

II. MATERIALS AND METHODS

This chapter outlined the design and development process of Circuit Breakers, detailing the data, approach, and steps taken to build the game and meet the study's objectives, including producing the final game and involving respondents in its evaluation.

A. Research Design

The project followed the Game Development Life Cycle 2.0 by Ramadan (2013), which was best segregated into four phases: Concept/Design, Alpha, Beta, and Release. Beginning with the game concept and planned features, the process advanced through asset creation and coding, iterative testing and evaluation, third-party beta testing for bug identification and feedback, and ultimately concluded with the game’s release once it was polished and ready for public deployment.

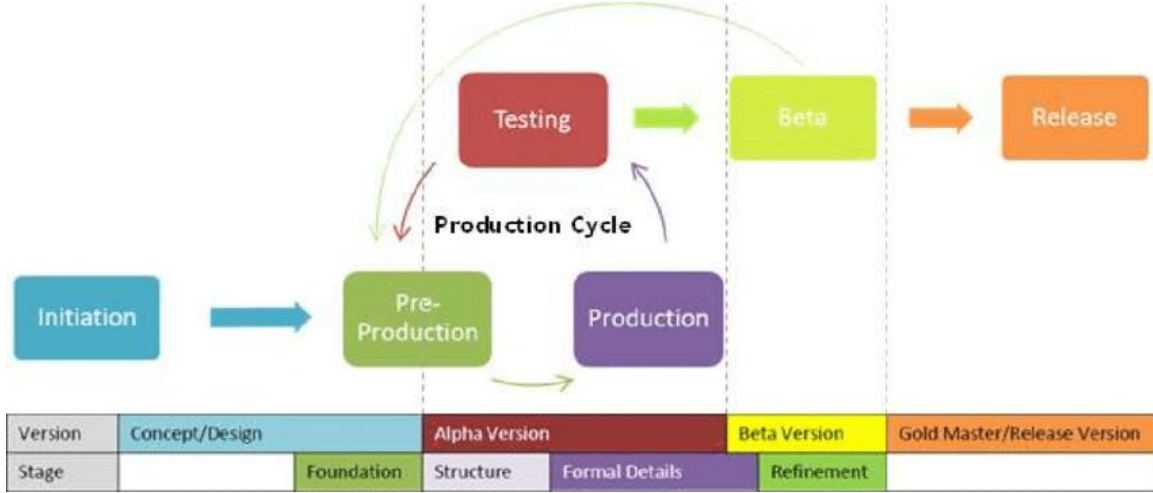


Figure 4: Game Development Life Cycle v2.0

B. Concept/Design Phase

The Concept/Design Phase involved defining Circuit Breakers’ objectives, learning features, gameplay mechanics, and technical requirements. This phase began with gathering and analyzing preliminary research data, including surveys, interviews, and expert consultations, to ensure the game effectively address gaps in digital literacy and cybersecurity awareness while remaining its replay ability and being engaging.

1. **Data Collection:** Preliminary surveys were sent to respondents through convenience sampling to assess their perceived knowledge of cybersecurity, engagement with traditional educational methods, and openness to game-based learning. An expert interview with cybersecurity educator Sir Emilio Manuel Macaso confirmed gaps in standardization and exposure to essential concepts. Findings highlighted key issues: limited understanding of basic cybersecurity principles, low retention of digital literacy content, and the need for engaging, strategy-based learning experiences. These insights validated the educational objectives of Circuit Breakers and informed its design priorities.
2. **Game Concept:** The game’s narrative centered on a player-controlled electronic device navigating a network infested with dynamic cybersecurity threats. Themes included cybersecurity, roguelike action, and educational engagement. The learning experience was integrated into the gameplay through:

- Threat Encyclopedia: A comprehensive library of threats (e.g., phishing, rootkits), their characteristics, and real-life examples that promoted awareness and prevention strategies.
- Item Encyclopedia: A library of defensive tools (e.g., firewalls, antivirus, 2FA) that the player could use strategically to protect their device.

Gameplay was structured around a single, procedurally generated virtual space, referred to as a “run” or “game,” where player progression was tracked through Gameplay Score (performance against threats) and Knowledge Score (strategic decisions and quiz completions). The roguelike loop emphasized strategy, adaptation, and consequence-driven learning, encouraging players to make decisions that mirrored real-world cybersecurity problem-solving.

3. *Requirement Analysis:* The requirement analysis confirmed the need for an engaging, analogy-based educational tool that bridged gaps in foundational cybersecurity knowledge. Key findings included:

- Limited Exposure: Students demonstrated minimal understanding of essential cybersecurity practices, such as avoiding personal information sharing online.
- Decreasing Engagement: Both students and experts emphasized the need for interactive, strategy-based learning to sustain attention and improve retention.
- Analogy-Based Learning: Linking in-game items and threats to real-world concepts reinforced memory and allowed for experiential understanding.

To address these requirements, the game incorporated adaptive threat spawning, consequence-driven feedback, and strategy-based decision-making to simulate realistic cybersecurity scenarios.

4. *Technical Requirements and Platforms:* The development of Circuit Breakers utilized a combination of software and platforms.

TABLE I
Circuit Breakers Software and Platform

Component	Software/Platform	Purpose
Front-End/Game Engine	Unity 2D	Built the core game, physics, graphics, and gameplay loop
Back-End/Data Management	Firebase	Collected real-time player data, handled authentication, and tracked progression
Code Framework	C#	Supported backend scripting and adaptive learning logic
2D Visual Assets	Libresprite	Created sprites, UI elements, and textures
Version Control	GitHub	Managed developer collaboration and version history

Audio Editing	Audacity	Edited and optimized background music and sound effects
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5. *Alignment with SDG Goals:* Circuit Breakers contributed to multiple Sustainable Development Goals.
SDG 4 – Quality Education: By integrating gameplay with cybersecurity learning, the game provided an innovative, engaging tool that enhanced digital literacy and understanding of threats.
SDG 8 – Decent Work and Economic Growth: The game promoted digital literacy, a key skill for employability, and inspired interest in cybersecurity careers.
SDG 9 – Industry, Innovation, and Infrastructure: Through serious game design and interactive pedagogy, the game demonstrated scalable, innovative educational infrastructure while fostering creativity and engagement.

C. Alpha Phase

The Alpha Phase consisted of the pre-production, production, and testing of game components and prototypes, which were continuously repeated until a working beta prototype was produced. This stage aimed to create, develop, and refine the game’s structure and mechanics to achieve the objectives of the study based on the established concept and design.

1. *System Design:* The system design of Circuit Breakers was developed using Unity 2D and C#, establishing the core gameplay framework that supported exploration, adaptive threat encounters, and analogy-based learning. This stage translated the outputs of the Concept/Design Phase into functional systems by implementing procedural generation, the roguelike progression loop, responsive item and threat interactions, and the integrated Knowledge and Gameplay scoring models. The design also incorporated UI/UX structures for the Threat and Item Encyclopedias, in-game quizzes, and the decision-based upgrade system, ensuring that gameplay, learning elements, and user navigation operated cohesively. These systems were built according to the identified requirements: embedding cybersecurity knowledge into gameplay, supporting player decision-making, and maintaining replay ability through dynamic threat spawning.
2. *Internal Testing:* Internal testing was conducted throughout early development to ensure that the foundational systems functioned as intended. During this phase, the development team tested core mechanics such as movement, enemy behavior, procedural generation stability, item functionality, and the triggering of quizzes and Knowledge Score updates. Early prototypes were played repeatedly to evaluate the accuracy of threat–item analogies, the responsiveness of controls, and the clarity of UI elements such as the encyclopedias. Issues detected during these sessions, such as inconsistencies in enemy spawning, imbalance in item effectiveness, UI misalignment, and quiz triggers failing to register, were corrected before progressing to the Beta Phase. This iterative testing ensured that the game’s learning and gameplay

systems operated smoothly and reflected the pedagogical goals established during the Concept/Design Phase.

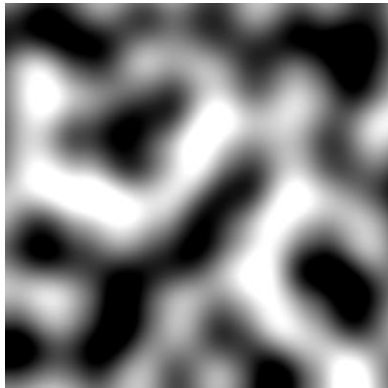
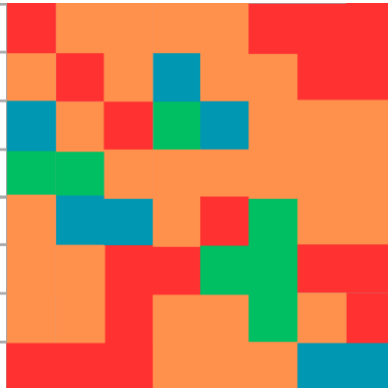
3. *Major Game Challenges*: The in-game challenges of Circuit Breakers reflected its core mechanics, learning goals, and roguelike structure from the following challenges.

- *Increasingly Difficult Threat Waves*: Players faced progressively harder waves of cybersecurity threats, as each horde increased in number, speed, variation, and attack patterns. This escalation required them to continually adjust their strategies to survive longer and adapt to rapidly changing scenarios.
- *Diverse Cybersecurity Threat Behaviors*: Each cybersecurity threat, such as Phishing, Ransomware, Adware, and Rootkits, demonstrated distinct movement patterns, attack effects, and status inflictions. Players needed to quickly recognize these behaviors and formulate appropriate responses, which made the experience both a gameplay and a learning challenge.
- *Strategic Item Selection and Build Optimization*: Throughout each run, players selected items and upgrades like Firewalls, Antivirus tools, 2FA, and Ad Blockers. Choosing the wrong combination could weaken their defense, while synergistic builds improve survival. Because threats were unpredictable, players had to make strategic decisions under uncertainty.
- *Managing Limited Resources*: Players worked with limited resources for purchasing items, upgrading defenses, and countering new threats. This constraint created impactful decision-making, as poor allocation often led to vulnerabilities and early defeat.
- *Adaptive Threat Spawning*: The game adjusted enemy compositions based on the player's chosen build. For instance, stronger Adware defenses often led to increased appearances of other threat types. Players had to avoid relying on a single strategy and instead anticipate how their choices influenced enemy behavior and predict future threats.
- *Quiz Challenges Under Gameplay Pressure*: Quizzes appeared at key intervals and contributed to the player's Knowledge Score. Players had to answer questions tied to real cybersecurity concepts while balancing the pressure of ongoing threats in order to receive a reward if they correctly answered the questions, while wary of the possibility of receiving a penalty if they answered incorrectly.
- *Completing Main Quests and Phase-Limited Quests*: Players also needed to complete main quests and time-restricted phase-limited quests that advanced the game's progression. These quests demanded efficient planning, quick decision-making, and prioritization under pressure, adding an additional layer of challenge alongside threat management to prevent bosses from being empowered upon quest failure.

- *Understanding the Threat and Item Encyclopedias*: The game provided encyclopedias explaining each threat and item, but players still needed to interpret and apply this information effectively. They were challenged to connect cybersecurity concepts to in-game mechanics and recall these details during fast-paced encounters.

4. *Algorithms Used*: The internal structure of Circuit Breakers was created using the following core algorithms and the scripts that they used.

TABLE II
Circuit Breakers Core Algorithms

Algorithms	Descriptions																																																																
Perlin Noise Map-Generation Algorithm	The game used a Perlin Noise-based algorithm to procedurally generate unique maps for every run. Perlin noise produced a gradient-based texture that served as the reference for tile placement, where noise values between 0.0 and 1.0 were scaled according to predefined tile weights representing different tiles. Tilesets and tile groups were prepared through functions that assigned ID codes to each tile type and organized them into their respective category folders. The weighted noise values were then converted into integers to form a digital terrain grid, which determined the specific tile spawned at each position. After the full grid was generated, the map was sliced into 8×8-tile chunks. This was handled by MapChunkGenerator.cs. These chunks dynamically loaded or unloaded based on their distance from the player, ensuring optimized performance throughout gameplay. This was handled by InfiniteMapManager.cs.																																																																
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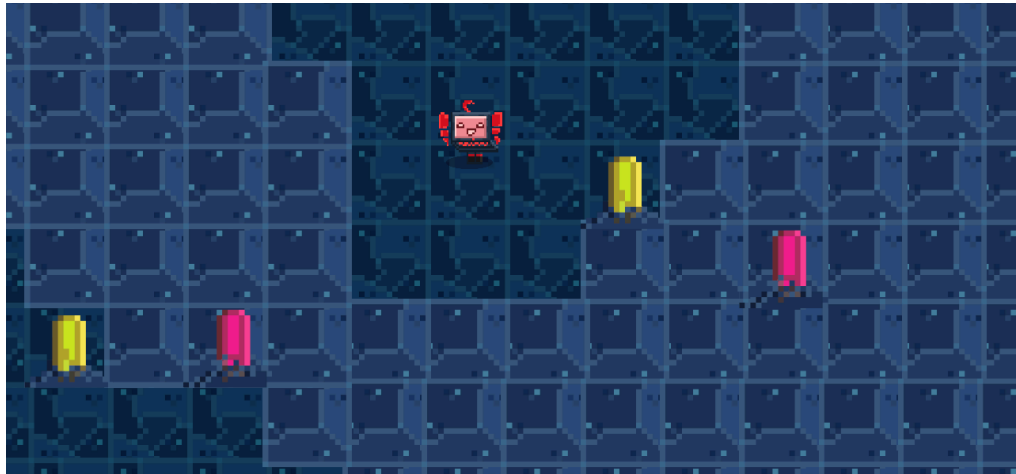


Figure 5: Map-Generation Process

Behavior Trees

Circuit Breakers implemented enemy behaviors using Behavior Trees, which structured AI decision-making through root, selector, sequence, and leaf nodes. Enemies spawned with default behaviors at the root node, while selector nodes allowed conditional decisions, sequence nodes executed consecutive actions, and leaf nodes contained specific behaviors or actions. Behavior Trees were also applied to player actions, including movement, status effects, and overload states. The implementation utilized Unity 6's built-in tools, including blackboard functions, to streamline AI behavior and decision-making. Sample implementations of scripted nodes include attack and movement scripts such as `BiteAttackAction.cs`, `FlurryStrikeAction.cs`, and `MovetoTargetAction.cs`.

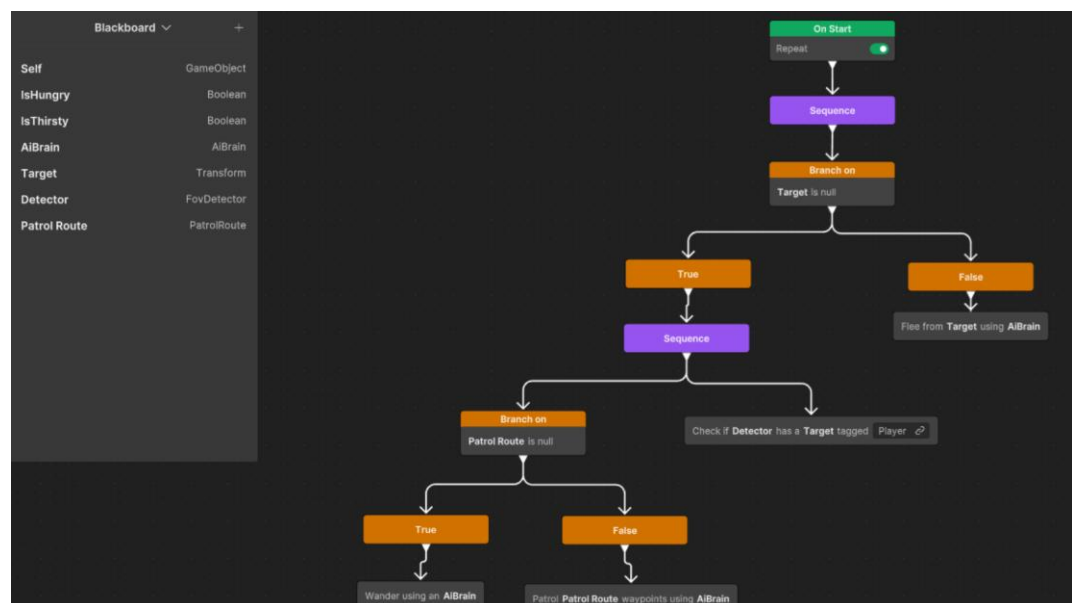
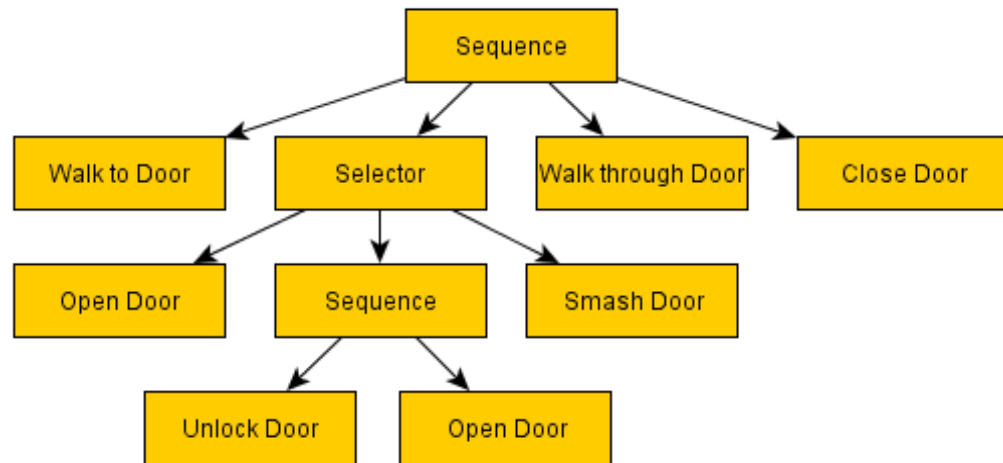


Figure 6: Behavior Tree Graph

Floating Text Feedback Response

The game incorporated a Floating Text Feedback system to provide dynamic, real-time guidance to players, highlighting item effectiveness and consequences of decisions through the FloatingText.cs script. Messages were displayed with configurable position, color, style, and animation, floating upward while shrinking and fading, and were efficiently managed through a pooling system. Together, these systems enhanced gameplay by creating responsive enemy AI and clear, immediate feedback that reinforced learning and strategic decision-making. This was managed by FloatingTextPool.cs, which implements object pooling to control performance by recycling game objects, using the primary object pooling script ObjectPoolManager.cs.



Figure 7: Floating Text Response Samples

Effectiveness Calculator

Circuit Breakers utilized an Effectiveness Calculator to determine how well items and weapons performed against specific enemy archetypes. The system loaded pre-defined effectiveness values from a CSV file using the KeywordEffectivenessMatrix.cs, which stored multipliers for individual item-enemy archetype pairings. The EffectivenessCalculator.cs then computed the effectiveness of any given weapon against single or multiple enemies, averaging results when applied to groups of threats. These calculations adjusted the player's damage output based on the weapon's effectiveness, categorizing it as Ineffective, Weak, Normal, Effective, or Super Effective. The outcomes were communicated to players via the Floating Text Feedback system, providing real-time guidance on the impact of their choices and reinforcing strategic decision-making during gameplay.

Item Name	# Malware (Infection)	# Phishing & Social Engineering	# Hacking & Evasion	# Zero-Day & Logic Bomb
2FA Token	0.75	0.75	1.25	2
AI Threat Prediction	0.75	0.75	1.25	1
AdBlock	0.5	1.75	0.75	0.5
Adware Remover	0.75	2	0.5	0.75
Antivirus	2	0.5	0.25	0.25
Antivirus Suite	1.75	0.75	0.5	0.5
Awareness Module	0.75	0.75	1.25	0.75
Backup Scheduler	0.75	0.75	0.75	1.25
Basic Firewall Ruleset	0.75	0.75	1.25	0.75
Battery Pack	1	1	1	1
Battery Pack	1	1	1	1
Compression Tool	1.25	0.75	0.75	0.75
Cooling Fan	1.25	0.5	0.75	0.5
CPU Cooler	1.25	0.75	0.75	0.75
CPU Overclock	1.25	0.5	0.75	0.5
Data Backup	0.75	0.75	0.75	1.5
Definition Update	1.25	0.75	0.75	0.75
Disk Defragmenter	0.5	1.25	0.75	0.5
Docking Station	0.75	1.25	0.75	0.75
Driver Update	0.75	1.25	0.75	0.75

Figure 8: Effectiveness CSV Table

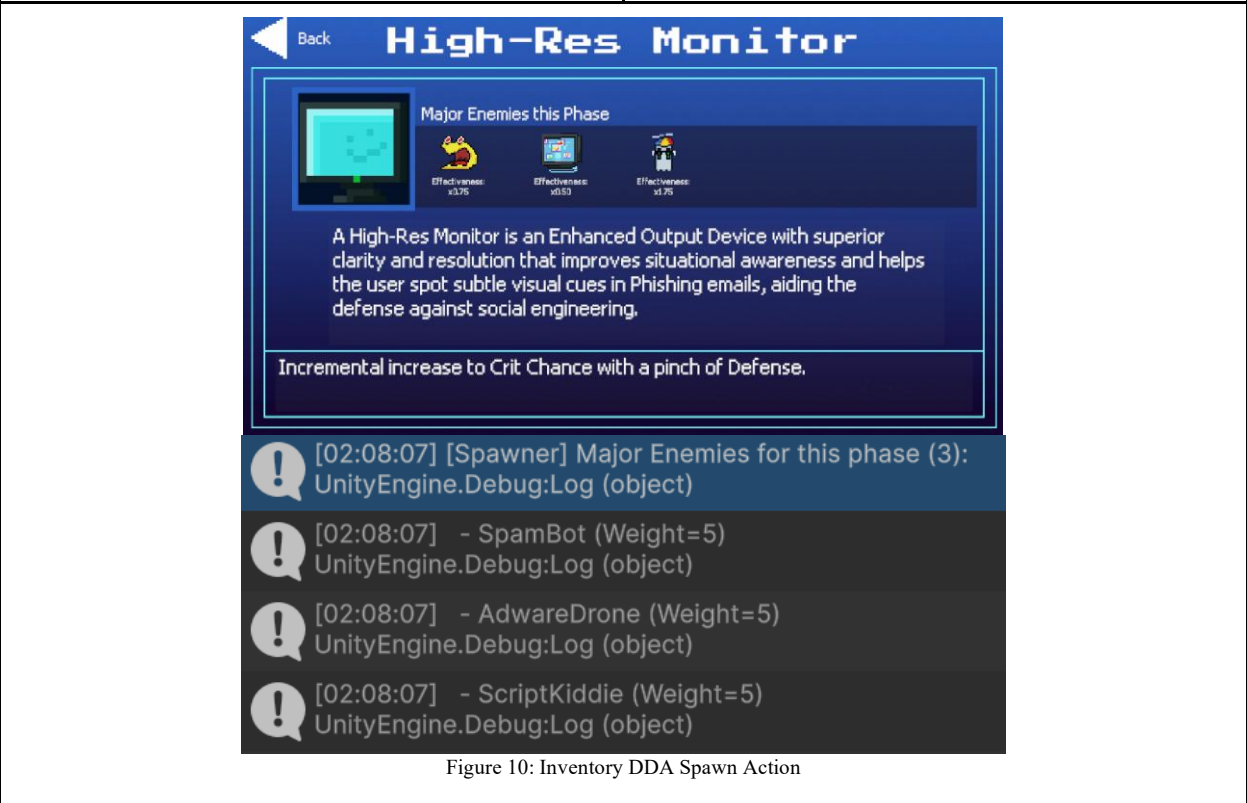


Figure 9: Effectiveness Calculation Samples

Inventory-Based DDA

Circuit Breakers implemented an Inventory-Based Dynamic Difficulty Adjustment system that adapted enemy spawns and challenge intensity based on the player's current items. The algorithm prioritized spawning enemies that were most resistant to the player's inventory, simulating the adaptive nature of real-world cyber threats. Each game phase calculated a set of MajorEnemies, which represented threats against which the player's items were least effective. The number of MajorEnemies varied according to the total potential enemies for the phase, ranging from one to five, and these were given higher spawn weights to influence gameplay difficulty. By dynamically adjusting enemy types and spawn rates in this way, the system ensured that players faced continuously engaging, strategic, and educational challenges that reinforced the need for careful planning and adaptation. This was primarily handled by EnemySpawner.cs, which utilized

	functions of the EffectivenessCalculator.cs and PlayerInventory.cs.
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Circuit Breakers implemented a Knowledge Profile algorithm to track, update, and calculate players' proficiency across four cybersecurity domains: Authentication, Privacy, Threat Detection, and Incident Response. Each domain maintained a current proficiency score and a ceiling value, which increased proportionally as the player advanced. Scores were scaled with a `BaseScoreMultiplier` and adjusted using a `BonusEfficiencyFactor` to provide diminishing returns, rewarding early progress while challenging players to achieve mastery. The Knowledge Profile algorithm continuously aggregated domain-specific scores to generate a total Knowledge Score, reflecting the player's overall understanding and performance. This system allowed the game to quantify learning outcomes, monitor progress in specific areas, and adapt challenges to reinforce retention of cybersecurity concepts. The calculations were handled by `KnowledgeProfile.cs` and were presented using `GameResultsModalScript.cs` at the end of each run.

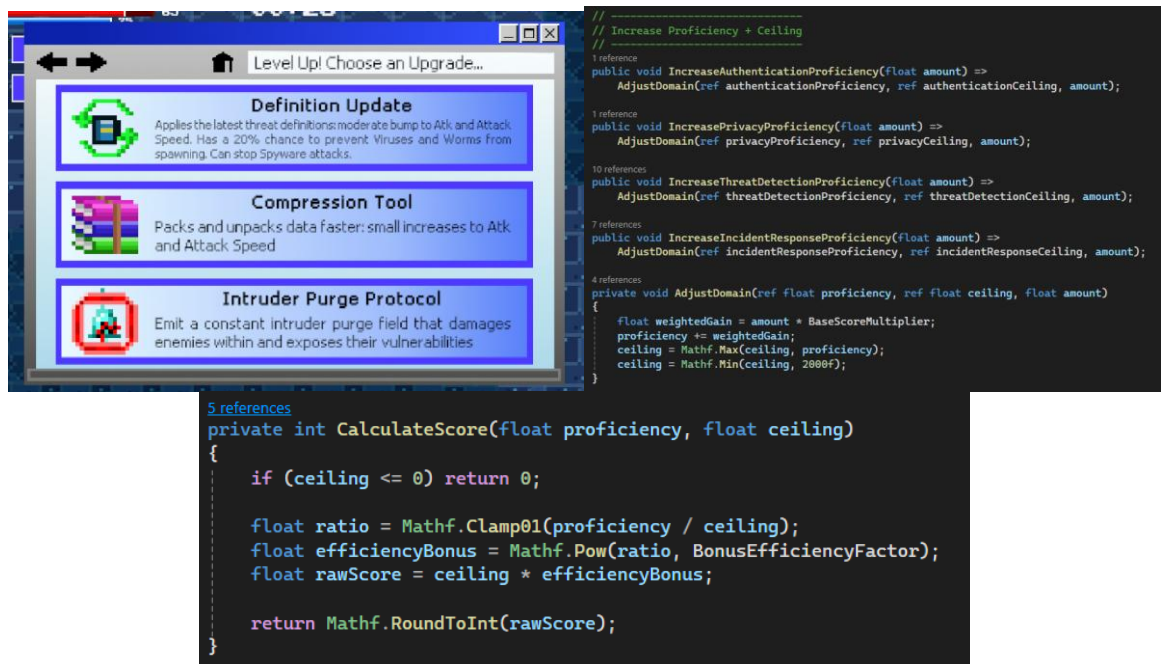


Figure 11: Knowledge Profile Calculation

Bonus Efficiency Factor Ratios

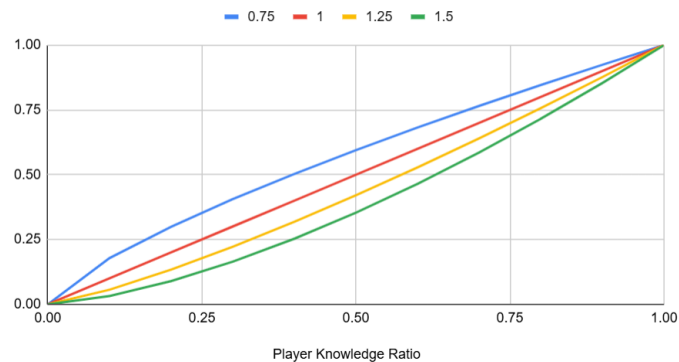
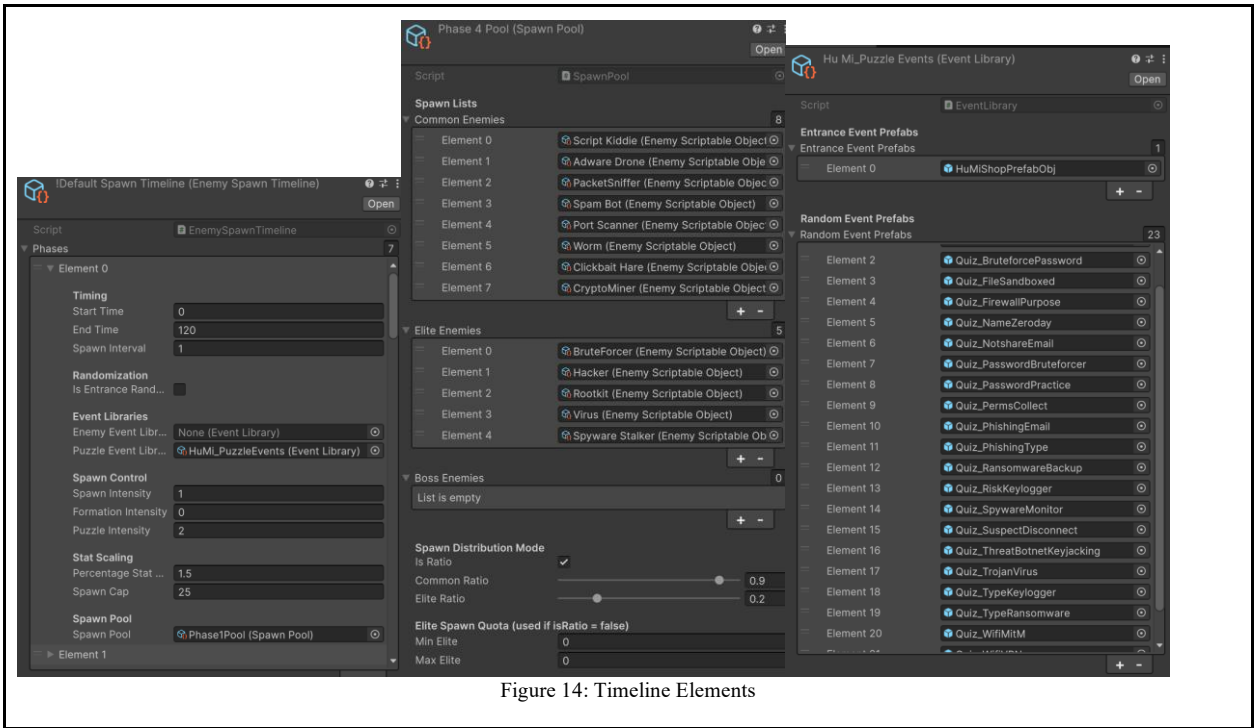


Figure 12: Bonus Efficiency Curve

Interactable Node

Circuit Breakers implemented an Interactable Node system to manage player interactions with key game objects, including Shop and Quiz Nodes. When players entered a node's range, a pop-up indicated the object was interactable, and leaving the range hid the prompt. Each node triggered a predefined event upon interaction, prioritizing Puzzle Events, Shop Events, or Quiz Events as applicable. The Shop Event algorithm supported item purchases, inventory upgrades, and detection of fake shops, influencing player scores in Threat Detection and Incident Response based on item effectiveness against major enemies. The Quiz Event algorithm generated randomized multiple-choice questions, rewarded Knowledge Points, and updated proficiency in relevant cybersecurity domains. InteractableNode.cs was the base script and utilized event scripts such as ShopEvent.cs and QuizEvent.cs to

	handle the core event logic, and each individual instances were spawned using ShopEventComponent.cs and QuizEventComponent.cs.
	
Figure 13: Interactable Node Pop-up	
Timeline Algorithm	Circuit Breakers employed a Timeline Algorithm to control the flow and pacing of each game phase. The algorithm defined phase durations, goals, enemy spawn frequency, intensity, limits, and scaling of enemy strength, as well as the timing of puzzle and event triggers. The system utilized scriptable objects to organize enemy pools and event libraries, ensuring that entrance events occurred at phase start while random events appeared throughout each phase under the EnemySpawnTimeline scriptable object. The TimelineController script managed phase transitions, triggered corresponding events, and adjusted gameplay progression dynamically. This approach allowed the game to regulate difficulty, maintain player engagement, and coordinate enemy and puzzle appearances in a structured yet adaptable manner.



5. *Art, UIUX, & Visual Design:* This section primarily presents a collection of figures illustrating the game's navigation flowchart and primary screen visuals. Circuit Breakers featured a clear and informative interface supporting gameplay, learning, and progression.

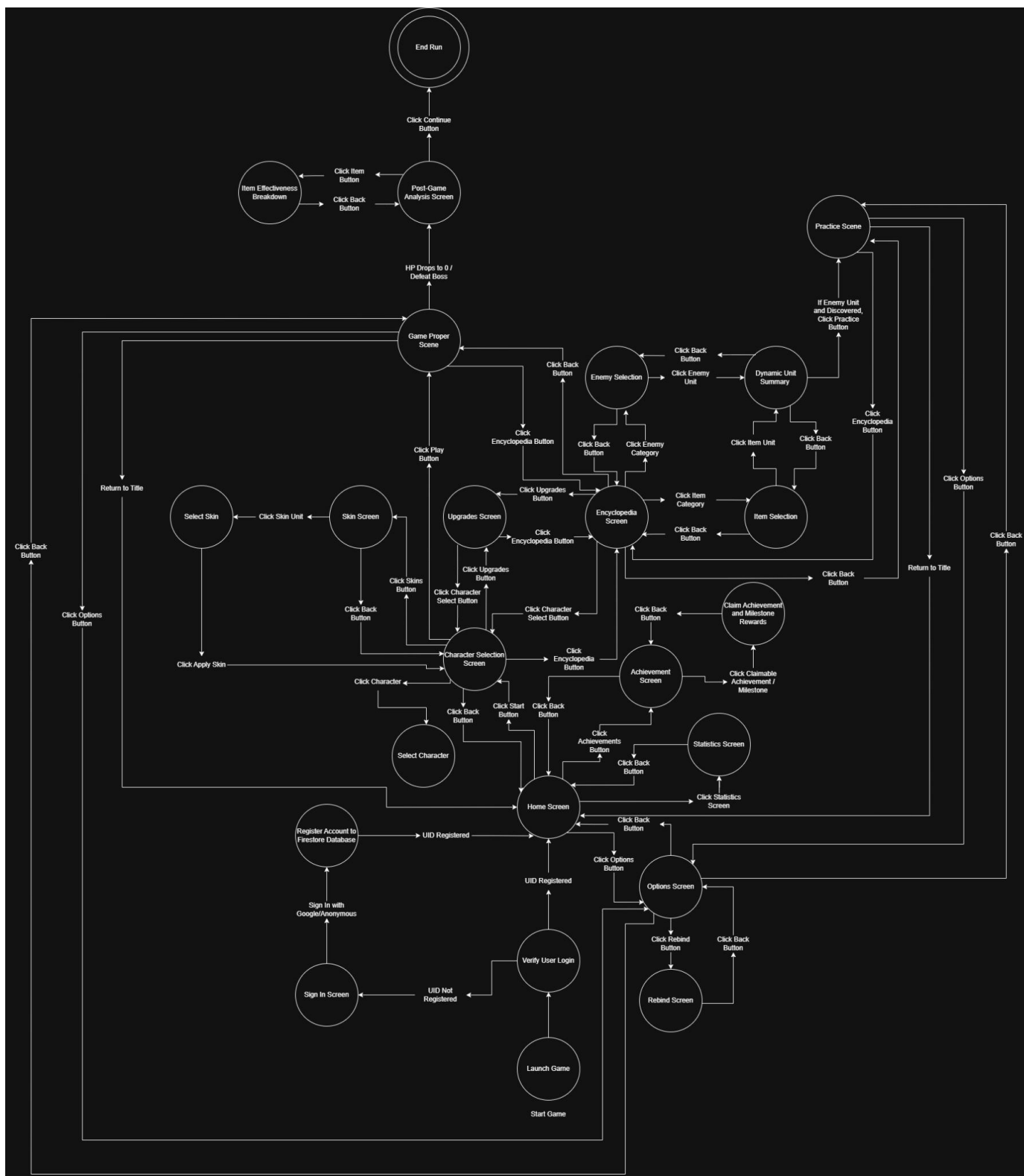


Figure 15: Circuit Breakers Finite State Navigation Flowchart



Figure 16: Circuit Breakers Gameplay



Figure 17: Circuit Breakers Encyclopedia Entry List



Figure 18: Circuit Breakers Encyclopedia Threat Entry



Figure 19: Circuit Breakers Local Sign-In Screen

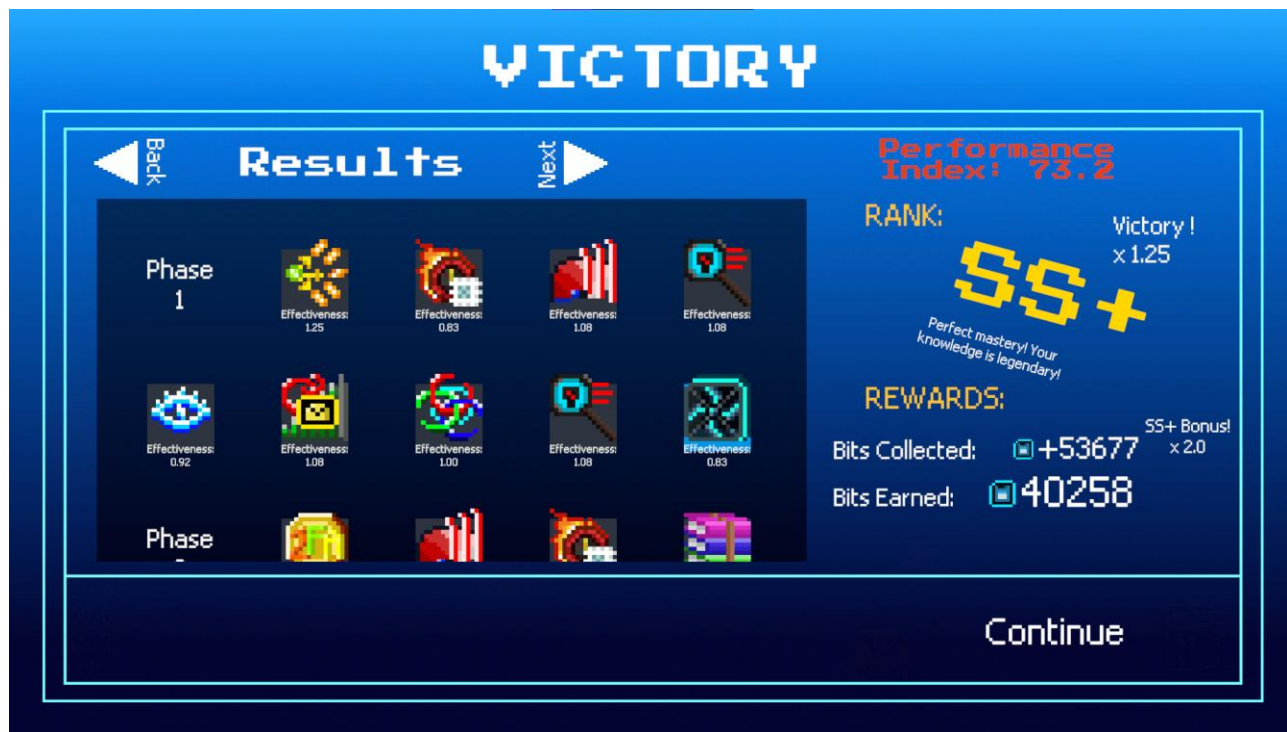


Figure 20: Circuit Breakers Post-Game Analysis 1

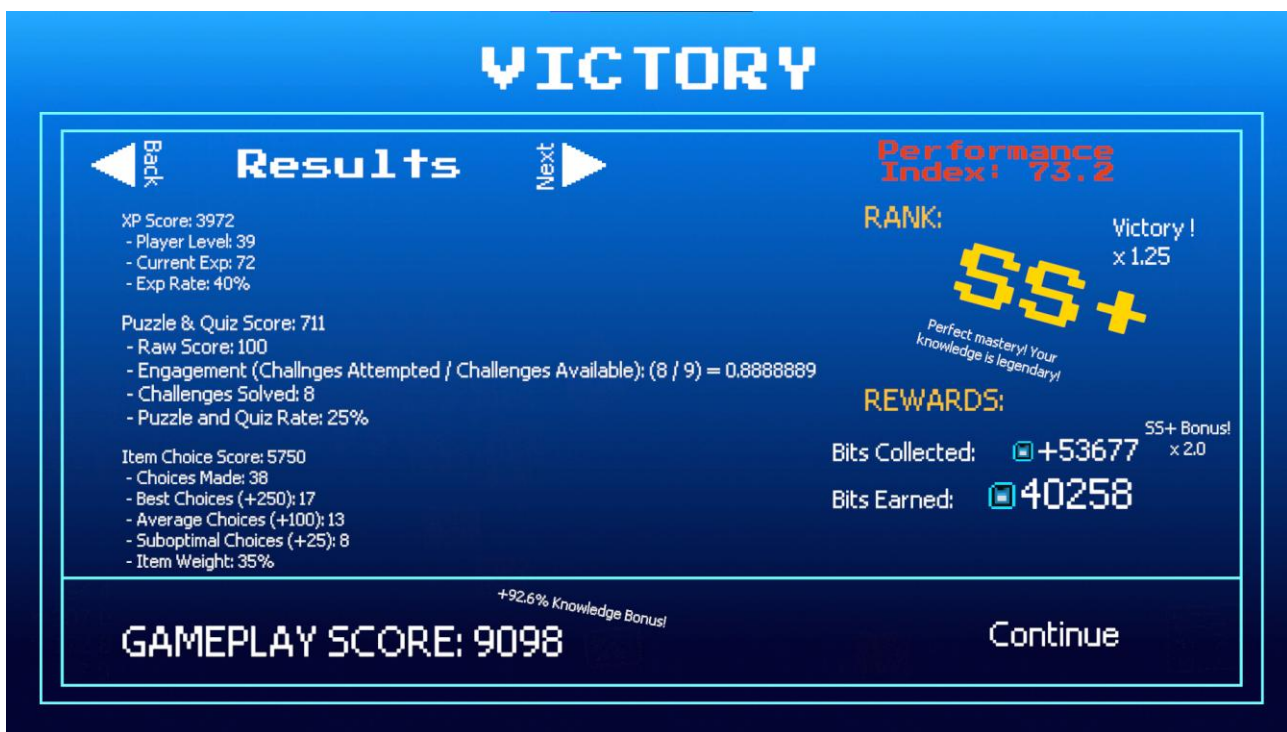


Figure 21: Circuit Breakers Post-Game Analysis 2

Circuit Breakers provided several secondary screens to support player learning and reflection. The Encyclopedia screen allowed players to access detailed information about threats and items, both during and outside gameplay, reinforcing understanding of discovered elements. The Statistics screen recorded and displayed the player’s session history and performance data. The Post-Game Analysis screen summarized the player’s score, reviewed item choices, and recorded each item’s effectiveness against major enemies in each phase, enabling players to evaluate the outcomes of their strategies and improve future runs.

D. Beta Phase

The Beta Phase focused on generating the study’s results through UAT testing and the collection of player data recorded in the Firebase database. This phase consisted of 31 respondents, including 17 participants aged 19–24, 10 participants aged 16–18, 3 participants aged 13–15, and 1 participant aged 25–34. Participants were selected through convenience sampling due to their availability, willingness to participate, and suitability as representatives of the general population rather than as members of a specialized group. During this phase, all player data were automatically recorded in the Firebase database, and each participant completed a structured UAT survey after playing the game.

E. Release Phase

The Release Phase was centered on the deployment and dissemination of Circuit Breakers. The game was prepared for launch on itch.io, a widely used platform for independent developers. As a desktop application, it was also suitable for potential release on larger distribution platforms such as Steam, allowing for broader accessibility and future scalability.

1. *Deployment Strategy*: The deployment strategy focused on preparing the game for release by meeting the technical and administrative requirements of the targeted platforms. These preparations included setting up developer accounts and generating a standalone build packaged as a ZIP file. These steps ensured that Circuit Breakers was ready for distribution on platforms such as itch.io or Steam, where future updates could have been delivered automatically to players.
2. *Localization Update*: For security, sustainability, and long-term maintainability, Circuit Breakers was updated to operate entirely offline. All player data was configured to run and save locally, eliminating reliance on third-party services such as Google or Firebase Database. This update ensured that the game functioned independently, reduced exposure to online threats, and preserved player privacy by preventing data transmission over the internet.

F. Participants, Sampling Plan, and Research Subjects

The study consisted of 51 respondents, comprising 40 participants aged 18–24, 6 participants aged 12–17, and 5 participants from other age ranges. Participants were selected through convenience sampling due to availability, willingness to participate, and representation of the general population, rather than involvement from a specialized group.

G. Data Analysis Procedure

The study consisted of 51 respondents selected through convenience sampling, chosen for their accessibility and willingness to participate. The sample primarily represented active digital users, with 40 participants aged 18–24, 6 aged 12–17, and 5 from other age groups. This composition allowed the study to assess general public interest in the game and baseline awareness of digital threats.

Data was collected in two phases: a pre-deployment survey and a post-deployment validation UAT phase. The initial survey, created and distributed through Google Forms, gathered statistics on User Interest and Motivation and Game Direction and User Expectations, including open-ended feedback that informed the game’s early design and feature requirements. After the deployment of Circuit Breakers, participants completed a structured UAT survey aligned with the study’s Objectives (O) and Problem Statements (P). The UAT assessed three areas:

1. *Demographics*: to establish baseline cybersecurity knowledge and gaming familiarity (P2).
2. *Learning & Digital Literacy*: to measure perceived knowledge gains, understanding of cybersecurity analogies, and motivational impact (P1, P2, O1, O2).
3. *Gameplay, Graphics, and Usability*: to evaluate technical execution, engagement, and the effectiveness of roguelike mechanics (P3, O3).

In parallel, all in-game data was automatically recorded in Firebase, providing objective metrics on player performance, progression, engagement, and quiz results. These metrics were cross-referenced with UAT feedback to validate the game’s effectiveness in enhancing cybersecurity awareness. The complete structure of the final UAT instrument was documented in the Final UAT Survey Breakdown, which detailed all questions and their alignment with the research goals.

III. RESULTS AND DISCUSSION

This chapter presents the comprehensive results derived from the multi-stage quality assurance process for Circuit Breakers, including formal Software Quality Assurance (SQA) testing and User Acceptance Testing (UAT).

A. Software Quality Assurance Testing

1. *Unit Testing Results*: Unit testing was conducted across nine (9) primary modules, totaling 102 test cases, to validate the integrity of individual functional components.

TABLE III

Summarized Unit Testing Results for Circuit Breakers

Modules	Test Cases	Passed	Failed	Passing Rate
---------	------------	--------	--------	--------------

M002: Main Menu	15	15	0	100%
M006: Options / Settings	12	12	0	100%
M013: Gameplay Scene (UI / Logic)	18	18	0	100%
M004: Achievements Screen	5	5	0	100%
M005: Statistics Screen	9	8	1	88.89%
M009: Permanent Upgrades / Shop	10	10	0	100%
M008: Skins Management	7	7	0	100%
M010/M011/M012: Encyclopedia System	14	14	0	100%
M014: In-Game Shop/Maintenance	12	10	2	83.33%
Total	102	99	3	97.06%

As shown in Table 1, the application demonstrated functional stability with 97.06% of all unit tests passing successfully. Failures were recorded in two modules: M005 (Statistics Screen) and M014 (In-Game Shop/Maintenance), primarily related to intermittent real-time data synchronization and currency handling during rejected transactions.

2. *Integration Testing Results:* Integration testing assessed the primary flow interactions and inter-module communication, totaling 21 test cases across seven flows. All integration flows achieved a 100% passing rate, confirming that inter-module communication and data handling via Firebase function correctly, guaranteeing the integrity and consistency of the game's core systems during transitions.

TABLE IV

Integration Testing Results

Modules	Test Cases	Passed	Failed	Passing Rate
M001→M003→M002: Auth & Alias Flow	3	3	0	100%

M002→M006: Main Menu to Options	2	2	0	100%
M002→M004/M005: Main Menu to Data Screens	4	4	0	100%
M002→M003: Main Menu to Run Preparation	3	3	0	100%
M003→M013: Run Preparation to Gameplay	4	4	0	100%
M013→M002: Gameplay End & Persistence	3	3	0	100%
M009/M010: Upgrade / Encyclopedia Access	2	2	0	100%
Total	21	21	0	100%

3. *System Testing Results:* System testing focused on end-to-end reliability across two primary modules: the Full Gameplay Loop (M023) and Run Saving and Data Persistence (M027).

Table V

System Testing Results

Modules	Test Cases	Passed	Fail	Passing Rate
Full Gameplay Loop (M023)	7	7	0	100%
Run Saving and Data Persistence (M027)	4	4	0	100%
Total	11	11	0	100%

System Testing achieved a perfect 100% passing rate, confirming that integrated modules function cohesively.

4. *Performance Testing Results:* This evaluation assessed the application's stability, efficiency, and resource utilization across four modules, totaling 14 test cases.

Table VI

Performance Testing Results

Modules	Test Cases	Passed	Fail	Passing Rate
---------	------------	--------	------	--------------

Game Performance (M024)	3	3	0	100%
Load Time and Resource Management (M025)	4	4	0	100%
Network Communication Performance (M026)	3	3	0	100%
PC Usability and Scaling (M028)	4	4	0	100%
Total	14	14	0	100%

5. *User Acceptance Testing (UAT) Results:* User Acceptance Testing (UAT) involved 31 participants to evaluate the game's usability, engagement, and efficacy.

Table VII

Number of UAT Testers According to Age Group

Age Range	Number of Participants (n)	Percentage (%)
13-15 years old	3	9.7%
16-18 years old	10	32.3%
19-24 years old	17	54.8%
25-34 years old	1	3.2%
Total	31	100.0%

6. *Tester Demographic:* The majority of participants were within the 19-24 years old cohort, providing robust usability data, while the inclusion of the target 13-18 age group ensured the core efficacy assessment remained focused on young learners.

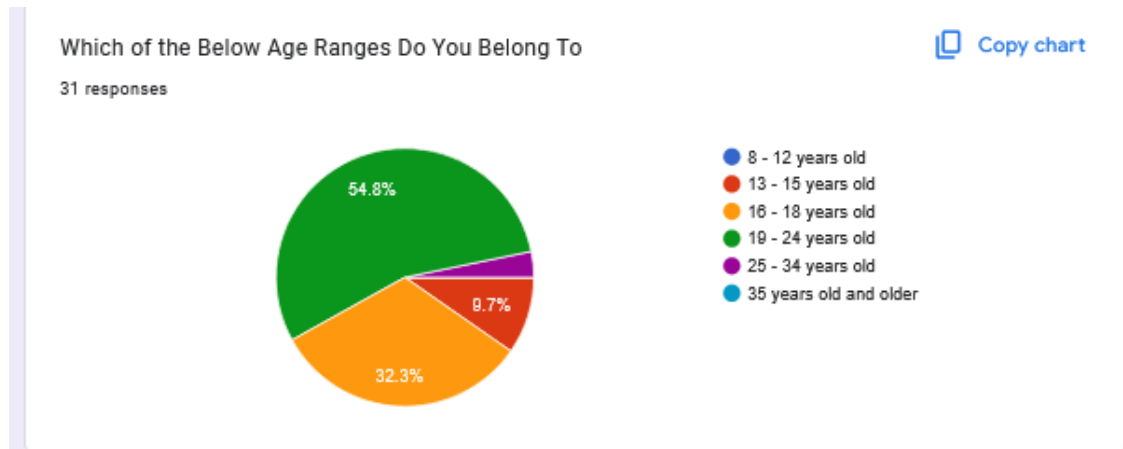


Figure 22: Circuit Breakers UAT Tester Age Range Pie Chart

B. Learning & Digital Literacy Quantitative Results

Table VIII

Testers' Feedback on the Learning Effectiveness of Circuit Breakers

Statements	Mean	Interpretation
I feel more confident identifying different types of digital threats.	4.19	Agree
I better understand the function of security tools.	4.26	Agree
The game showed consequences of cybersecurity mistakes.	4.55	Strongly Agree
Mechanics represent real-world concepts.	4.45	Agree
I would recommend this game for digital literacy learning.	4.77	Strongly Agree
Compared to traditional learning, this was more engaging.	4.61	Strongly Agree
Strategic choices helped me remember functions.	4.48	Agree

C. Learning & Digital Literacy Qualitative Responses

Qualitative data supported the high quantitative scores, validating the use of analogical representation in the game's design, particularly for the Worm and Adware/Adblocker concepts. Conversely, abstract concepts such as "Man in the Middle (MiTM)" were noted as needing clearer representation.

D. Gameplay, Graphics, and Usability Quantitative Results

Table IX

Testers' Feedback on Gameplay, Graphics, and Usability

Statements	Mean	Interpretation
The core gameplay loop was fun and engaging.	4.71	Strongly Agree
The visual aesthetic was appealing and cohesive.	4.52	Strongly Agree
Choices felt meaningful.	4.55	Strongly Agree
Replay value.	4.58	Strongly Agree
Motivation after losing.	4.48	Agree
Ease of navigation.	4.42	Agree
Thinking about effective purchases.	4.39	Agree
Difficulty was fair.	4.10	Agree
Detecting fake HuMi Shop.	3.39	Neutral

The highest scores confirmed strong player satisfaction, while the neutral score for detecting the fake HuMi Shop ($\bar{x} = 3.39$) indicates a need for refinement in the phishing analogy's visual cues.

E. Discussion of Results

The UAT results provide robust statistical and contextual validation that the design successfully countered the core educational problems identified in the thesis.

Table X

Alignment with Problem Statements

Problem Statement	Summary of Validation	Key Data
P1: Decreasing Engagement	Decisively Solved. The gamified, roguelike structure is overwhelmingly viewed as a superior and highly motivating alternative to conventional teaching methods. The game's inherent fun factor drives sustained exposure to the educational content.	4.61 (Strongly Agree) that the game is significantly more engaging than traditional learning. 4.71 (Strongly Agree) that the core gameplay loop was fun.

P2: Limited Exposure & Lack of Analogy	Successfully Addressed. The abstract analogies used to represent cybersecurity concepts are highly effective. Testers confirmed the mechanics clearly translate to real-world security functions, overcoming the challenge of teaching non-literal concepts.	4.45 (Agree) that game mechanics clearly represent real-world concepts. Qualitative responses lauded the Worm and Adware/Adblocker as perfect functional analogies.
P3: Lack of Modern Methods/Retention	Highly Validated. The need to make strategic, context-dependent choices within the game directly enforces knowledge application, which is crucial for retention and practical digital literacy.	4.48 (Agree) that strategic choices helped players better remember item functions. Qualitative data confirms a direct, positive behavioral change (e.g., installing ad blockers).

The UAT strongly validates the hypothesis that the roguelike GBL approach effectively solved the problems of disengagement and poor knowledge retention in cybersecurity education by offering a highly engaging, functionally analogous, and strategically demanding learning environment.

The feedback confirms the successful integration of learning mechanisms into the core gameplay loop, fulfilling the project's primary design goals.

Table XI

Fulfillment of Core Objectives

Objective	Fulfillment Summary	Key Data & Insight
O1: Design Game as an Alternative Means of Education	Fulfilled (with refinement needed). The game successfully serves as an alternative educational tool, evidenced by improved player confidence in identifying threats and understanding security functions. However, abstract concepts require better visual differentiation.	4.37 (Agree) that in-game descriptions were helpful and clear. Refinement: The representation of Man in the Middle (MiTM) and Cryptominer were cited as confusing, requiring clearer mechanics or descriptions.
O2: Strategic Challenges and Decision-Making	Fulfilled (with a critical gap). The game design successfully uses consequence reinforcement to teach responsible behavior, scoring highly on showing the effects of mistakes and promoting strategic purchasing.	4.55 (Strongly Agree) that the game effectively showed the consequences of mistakes. Critical Gap: The Phishing Analogy (fake shop) scored 3.39 (Neutral), indicating the visual cues for threat identification are insufficient for proactive learning.
O3: Engaging Roguelike Gameplay	Exceeded. The choice of the roguelike genre is unequivocally validated, creating a compelling loop that encourages repeated play and maximizes exposure to educational content.	4.71 (Strongly Agree) for the core gameplay loop being fun. 4.48 (Agree) that losing a run was motivating, confirming the success of the genre's key replayability mechanism.

The UAT confirms that the strategic and educational goals of the game are being met through its roguelike mechanics, although visual and structural refinements are necessary to close the gap on teaching complex or visually subtle threats like phishing. In addition to the respondents from the UAT, we also consulted with a cybersecurity expert and educator regarding the overall integration between gameplay and education, as well as their opinion on parts that could be improved. Feedback primarily revolved around the overall complexity of the terms used as well as the difficulty of the quiz questions. Players unfamiliar with cybersecurity may not know certain concepts mentioned that would be a prerequisite to understanding certain items or questions, which could hamper the game's teaching capabilities. Their recommendation involved allowing the player to choose the baseline of their knowledge, letting them input their age and proficiency with cybersecurity, and scaling the difficulty and complexity from there. However, overall response to the gameplay was positive, noting how the gameplay loop and visuals encouraged players to go back for more runs even after failing, which can help facilitate learning through repetition.

1. *Causal Analysis (Addressing the Problem Statements)*: The convergence of subjective UAT scores and objective Firebase data provides conclusive evidence regarding the game's educational efficacy and engagement.
 - *P1: Decreasing Engagement in Cybersecurity Education*. This problem was solved by the roguelike design. The core loop's high subjective rating ($\bar{x} = 4.71$) and the objective, aggregate Total Play Time of 4,180.57 minutes (approximately 70 hours) across 29 testers provided empirical proof of sustained, unforced engagement, which was a prerequisite for successful Game-Based Learning (GBL). The score of 4.48 (Agree) for motivation after losing directly validated the roguelike mechanic for driving replayability and maximum content exposure.
 - *P2: Limited Exposure and Lack of Functional Analogy*. The high agreement that mechanics clearly represent real-world concepts ($\bar{x} = 4.45$) validated the success of analogical design. Furthermore, self-directed learning was confirmed by the time spent in the Circupedia (up to 53.48 minutes by top players), which indicated that the desire for in-game success effectively pulled players toward educational content.
 - *P3: Lack of Modern Methods and Poor Retention*. The hypothesis that application-enforced learning improves retention was confirmed. The strategic complexity and reward-based knowledge gateway (quizzes) forced a rapid Learn-Test-Apply cycle. This was evidenced by the 100% Quiz Accuracy Rate among the most engaged players, demonstrating that knowledge was successfully acquired and applied for strategic mastery. Crucially, qualitative data confirmed a direct transfer of learning to real-world behavior (e.g., users installing real-world adblockers).
2. *Validation of Core Objectives*: The UAT results confirmed the successful integration of learning mechanisms into the core gameplay loop, directly fulfilling the project's primary design goals.
 - *O1: Design Game as an Alternative Means of Education*. This objective was largely fulfilled, evidenced by improved player confidence in identifying threats ($\bar{x} = 4.19$) and enhanced understanding of security functions ($\bar{x} = 4.26$). The high score for the clarity of textual learning aids ($\bar{x} = 4.37$) confirmed that the text-based resources were successful. However, the need for clearer visual and mechanical representations for abstract concepts (like Man in the Middle) suggested that refinement was needed for the quality of analogy across all threats.

- O2: Strategic Challenges and Decision-Making. This objective was highly successful in its core goal of reinforcing responsible behavior. The highest learning-specific score was $\bar{x} = 4.55$ (Strongly Agree) for effectively showing the consequences of common cybersecurity mistakes. The critical gap remained the Phishing Analogy (fake shop), which scored $\bar{x} = 3.39$ (Neutral), demonstrating a failure in teaching proactive threat identification before interaction.
- O3: Engaging Roguelike Gameplay. This objective was unequivocally validated, confirming the genre's ability to create a compelling, repeatable loop essential for sustained learning exposure. The core gameplay loop was rated $\bar{x} = 4.71$ for fun, and the motivational effect of losing a run scored $\bar{x} = 4.48$. This confirmed the design successfully drove replay intention, ensuring players remained in the learning environment long enough to internalize the educational content.

IV. CONCLUSION AND RECOMMENDATIONS

A. Conclusion

Circuit Breakers validated the hypothesis that the roguelike genre provided a superior, highly engaging, and pedagogically sound framework for Game-Based Learning in cybersecurity. The design was a powerful counter to educational disengagement (P1), proving that players committed significant, unforced time, as evidenced by the aggregate 4,180.57 minutes of playtime, to a challenge that rewarded knowledge. The project demonstrated that strategic game mechanics effectively supplemented learning, turning the pursuit of game mastery into a more engaging avenue for the acquisition of domain knowledge. The core loop's success in enforcing consequence-based learning (O2) and in translating abstract threats into functional analogies (O1) firmly established the game as a viable and effective educational supplement.

B. Limitations and Recommendations for Future Development

1. *Critical Accessibility and Scaling Limitation:* The expert interview identified the most significant barrier to broad educational adoption: the current static difficulty and technical complexity of the educational content. Future development must incorporate Adaptive Learning Scaling based on an initial user assessment (age/proficiency) to dynamically adjust content.
 - Quiz Difficulty: Easier, foundational questions for younger/novice players.
 - Encyclopedia Descriptions: Descriptions needed to be simplified to be "less technical and more for kids" at lower levels, and progressively increase in depth as the player's proficiency grew.
2. *Asynchronous Software Update Limitation:* The standalone game of Circuit Breakers, directly exported from Unity, by itself cannot update all copies of the game that players may have. This implies that cybersecurity concepts, such as malware definition and security tools, may become obsolete unless the user directly updates the game if the game releases a new version. Assuming that the game is regularly updated, it is best to distribute Circuit Breakers to major online platforms for distributing and playing video games, such as Steam and itch.io. This would allow players to keep their versions of Circuit Breakers up-to-date and for the educational aspect of the game to remain relevant.
3. *Recommendations for Future Research:* The following recommendations identified new areas of research based on the findings of this study, ensuring the continued academic significance of the validated GBL model.

- *Adaptive Learning Scaling Research*: While the core GBL model was validated, the expert interview identified a critical limitation related to the initial technical complexity of the educational content. Future research should explore the feasibility and efficacy of incorporating an initial user assessment (age/proficiency) to dynamically scale Quiz Difficulty and Encyclopedia Descriptions. This was essential for the model to achieve true educational supplement status across the broad, non-STEM target audience of young learners.
- *Longitudinal Transfer of Learning Assessment*: Future work should focus on longitudinal studies to quantify the long-term retention of cybersecurity knowledge acquired through Circuit Breakers compared to traditional classroom methods. This would have involved pre- and post-testing that measured the direct transfer of learned concepts into real-world defensive behaviors, thereby further quantifying the behavioral efficacy of the GBL model and its significance as a new method of instruction.

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